

TIRUCHIRAPPALLI, India

WMO: 433440

Lat: 10.765N

Lon: 78.710E

Elev: 88

StdP: 100.27

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.0	20.9	14.8	10.6	30.9	15.9	11.4	29.9	8.5	27.8	8.2	27.8	1.9	20	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.5	39.2	26.0	38.8	25.9	37.9	25.7	27.7	35.0	27.2	34.2	26.9	33.8	3.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	30.0	25.8	21.4	29.7	25.2	20.6	29.2	89.3	35.6	87.1	34.5	85.5	34.1	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.1	8.2	DB	18.3	40.7	0.9	1.0	17.6	41.4	17.1	42.0	16.6	42.5	15.9	43.2	(4)
(5)			WB	16.7	29.2	1.0	1.4	16.0	30.2	15.4	31.0	14.9	31.8	14.2	32.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	25.9	27.6	30.0	32.2	32.4	32.1	31.7	31.0	30.4	28.9	27.0	25.8	(6)	
	DBStd	2.76	1.07	1.33	1.59	1.63	1.75	1.32	1.19	1.28	1.38	1.65	1.37	1.17	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	7152	493	493	621	666	695	662	672	651	613	586	509	491	(10)	
	CDD18.3	4111	235	259	363	416	437	412	413	393	363	328	259	232	(11)	
	CDH23.3	50743	1925	2677	4619	5957	6297	5811	5718	5251	4673	3694	2344	1777	(12)	
	CDH26.7	26744	667	1234	2530	3616	3882	3438	3277	2857	2422	1583	737	501	(13)	
(14) Wind	WSAvg	3.8	3.5	3.3	2.8	2.6	3.7	5.2	6.0	5.0	3.8	2.8	3.1	3.6	(14)	
(15) Precipitation	PrecAvg	915	18	11	9	35	59	36	62	87	170	184	138	104	(15)	
	PrecMax	1314	172	125	63	160	159	122	202	220	415	347	329	482	(16)	
	PrecMin	483	0	0	0	0	4	0	1	10	49	46	4	8	(17)	
	PrecStd	206	40	26	15	40	38	35	55	60	82	86	91	105	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	35.8	38.9	40.2	40.9	39.8	39.0	38.1	38.0	36.9	33.2	32.1	(19)	
		MCWB	23.2	23.9	24.4	26.2	26.5	26.2	25.2	25.2	25.2	25.3	24.9	23.8	(20)	
	2%	DB	31.6	34.2	37.6	39.2	39.5	38.2	37.9	37.1	36.9	35.2	32.2	31.0	(21)	
		MCWB	22.9	23.3	24.6	26.0	26.4	25.9	25.2	25.2	25.0	25.5	24.6	23.7	(22)	
	5%	DB	30.9	33.2	36.6	38.2	38.8	37.4	37.0	36.2	35.9	34.1	31.2	30.1	(23)	
		MCWB	22.7	23.1	24.4	25.8	26.3	25.7	25.1	25.1	25.0	25.4	24.5	23.4	(24)	
	10%	DB	30.0	32.2	35.2	37.2	37.7	36.5	35.9	35.2	34.9	32.9	30.2	29.2	(25)	
		MCWB	22.5	23.0	24.2	25.8	26.3	25.5	25.2	25.0	25.1	25.3	24.4	23.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	25.9	27.2	28.4	28.5	28.2	27.6	27.1	27.2	26.8	26.6	26.0	(27)	
		MCDB	29.6	31.4	33.3	35.8	35.9	36.4	35.2	33.5	32.6	31.7	30.2	29.1	(28)	
	2%	WB	24.6	25.1	26.4	27.6	27.7	27.2	26.7	26.5	26.6	26.5	26.1	25.3	(29)	
		MCDB	28.9	30.7	32.7	34.6	35.3	35.1	34.2	33.3	32.4	31.3	29.6	28.5	(30)	
	5%	WB	24.1	24.6	25.9	27.1	27.2	26.6	26.2	26.1	26.2	26.2	25.7	24.8	(31)	
		MCDB	28.1	30.0	32.3	33.7	34.6	34.5	33.6	33.0	32.0	30.8	29.1	27.9	(32)	
	10%	WB	23.6	24.2	25.5	26.7	26.9	26.1	25.7	25.7	25.9	25.9	25.3	24.3	(33)	
		MCDB	27.7	29.4	31.6	33.2	34.2	33.9	33.2	32.5	31.7	30.5	28.5	27.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.5	11.4	10.7	10.5	9.6	9.2	9.1	9.2	7.9	6.7	7.4	(35)	
		MCDBR	9.8	11.3	12.3	11.8	11.6	10.7	10.3	10.2	10.4	9.6	8.3	8.5	(36)	
	5% WB	MCWBR	3.0	3.0	2.9	2.8	3.2	3.1	2.9	3.0	3.2	3.0	2.7	3.0	(37)	
		MCWBR	8.0	9.6	10.8	10.7	10.7	10.3	9.5	9.4	9.3	8.2	6.7	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.458	0.457	0.503	0.524	0.572	0.488	0.505	0.494	0.482	0.529	0.507	0.486	(40)	
		taud	2.192	2.190	2.067	2.057	1.935	2.206	2.147	2.186	2.222	2.068	2.112	2.137	(41)	
	Ebn at Noon		847	863	828	802	749	805	795	815	833	789	796	809	(42)	
		Edh at Noon	145	150	173	173	191	144	154	150	146	168	156	151	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.01	5.87	6.29	6.24	5.99	5.62	5.45	5.56	5.59	4.83	4.16	4.22	(44)	
	RadStd		0.24	0.25	0.35	0.24	0.34	0.31	0.28	0.27	0.29	0.29	0.41	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	-0.16	N/A	+1.01	N/A	-0.30	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.36	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page